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| Inventor(s)          | Sulai; Yusufu Njoni Bamaxam et al. |

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### Enabling eye tracking in pancake lens optics

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#### Abstract

Ghost images can interfere with eye tracking in a system that uses folded optics, such as is an artificial-reality display. Optical elements, such as waveplates and/or polarizers, can be used to attenuate or eliminate light causing the ghost images.

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**Inventors:** Sulai; Yusufu Njoni Bamaxam (Snohomish, WA), Lam; Wai Sze Tiffany (Lynnwood, WA), Peng; Fenglin (Redmond, WA), Gollier; Jacques (Sammamish, WA)

**Applicant:** Meta Platforms Technologies, LLC (Menlo Park, CA)

**Family ID:** 1000006359312

**Assignee:** Meta Platforms Technologies, LLC (Menlo Park, CA)

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*Primary Examiner:* Shafer; Ricky D

*Attorney, Agent or Firm:* Greenberg Traurig, LLP

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. Non-Provisional application Ser. No. 16/178,177, filed Nov. 1, 2018, titled “ENABLING EYE TRACKING IN PANCAKE LENS OPTICS,” which is herein incorporated by reference in its entirety for all purposes.

## BACKGROUND

(1) Artificial reality is a form of reality that has been adjusted in some manner before presentation to a user, which may include, e.g., virtual reality (VR), augmented reality (AR), mixed reality (MR), hybrid reality, or some combination and/or derivatives thereof. In a virtual reality (VR) system, a near eye display panel can be configured to present virtual reality images that depict objects in a virtual environment. The display panel may combine real objects with virtual objects, as in AR or MR applications.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Illustrative embodiments are described with reference to the following figures.
- (2) FIG. 1 illustrates a ray tracing diagram of an embodiment of an artificial-reality display with a lens arrangement for a pancake lens.
- (3) FIG. 2 depicts a schematic of an optical path of an embodiment of the lens arrangement.
- (4) FIG. 3 is a picture of ghost images from a camera used for eye tracking with the lens arrangement.
- (5) FIG. 4 illustrates a ray tracing diagram of an embodiment of a camera used for eye tracking with the lens arrangement.
- (6) FIG. 5 illustrates a ray tracing diagram of an embodiment of a partial mirror with a dielectric coating.
- (7) FIG. 6 illustrates a ray tracing diagram of an embodiment of an eye-tracking system with a circular reflective polarizer.
- (8) FIG. 7 illustrates a ray tracing diagram of an embodiment of an eye-tracking system for the lens arrangement.
- (9) FIG. 8 depicts a schematic of optical paths of an eye-tracking system with the lens arrangement.
- (10) FIG. 9 is a picture of a *Narcissus* effect of a camera.
- (11) FIG. 10 depicts a schematic of an optical path of an eye-tracking system for reducing the *Narcissus* effect.
- (12) FIG. 11 illustrates an embodiment of a flowchart of a process for using an eye-tracking system with a pancake lens.
- (13) The figures depict embodiments of the present disclosure for purposes of illustration only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated may be employed without departing from the principles, or benefits touted, of this disclosure.
- (14) In the appended figures, similar components and/or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

### DETAILED DESCRIPTION

- (15) In the following description, for the purposes of explanation, specific details are set forth in order to provide a thorough understanding of certain inventive embodiments. However, it will be apparent that various embodiments may be practiced without these specific details. The figures and description are not intended to be restrictive.
- (16) The present disclosure relates generally to folded optics. More specifically, and without limitation, this disclosure relates to eye tracking in a head-mounted display while using a pancake lens. A pancake lens is a lens or lens system that uses mirrors to fold an optical path. In some

virtual-reality (VR) systems, in which a pancake lens isn't used, a hot mirror (an optical element that transmits visible light and reflects infrared light) is used to gather reflected light from an eye for eye tracking. To reduce volume of a head-mounted display, a pancake lens is used in an artificial-reality display. In some configurations, a pancake lens comprises a partial mirror, a quarter-wave plate, and a reflective polarizer. A pancake lens has more surfaces, which can cause one or more ghost images imaged by a camera used for eye tracking.

(17) There are several possible solutions to reduce ghost images for eye tracking while using a pancake lens. A first solution is to use a dielectric coating for the partial mirror of the pancake lens, which does not reflect in infrared. A second solution is to use a reflector having a narrow band circular reflective polarizer (e.g., a cholesteric liquid crystal). A third solution is to add, in addition to the pancake lens, a first quarter-wave plate, a second quarter-wave plate, and a linear polarizer (e.g., an absorptive linear polarizer). The first quarter-wave plate is placed between the partial mirror of the pancake lens and a hot mirror. The second quarter-wave plate is placed between the hot mirror and a camera used for eye tracking. The linear polarizer is placed between the second quarter-wave plate and the camera. In some embodiments, a third quarter-wave plate is placed between the linear polarizer and the camera to reduce reflections off a lens of the camera (e.g., a fast axis of the third quarter-wave plate is at 45 degrees to the transmission axis of the linear polarizer).

(18) FIG. 1 illustrates a ray tracing diagram of an embodiment of an artificial-reality display **100** (e.g., for a virtual-reality system) with a lens arrangement **102**. The lens arrangement **102** is sometimes referred to as a pancake lens. Light from a display **104** is transmitted through the lens arrangement **102** to an eye box **108**. The lens arrangement **102** expands image light from the display **104**. The lens arrangement **102** comprises a first mirror M-1 and a second mirror M-2. The first mirror M-1 and the second mirror M-2 fold light onto itself. The first mirror M-1 is a partial reflector **112**. The partial reflector **112** has a transmission T, wherein T is equal to or greater than 20% or 40% and equal to or less than 60% or 80% (e.g., the partial reflector **112** is a 50/50 mirror with  $T=50\% \pm 2, 5, \text{ or } 10\%$ ). The first mirror M-1 is on a curved surface of a lens (e.g., for focusing light from the display to an eye of a user). The second mirror M-2 is a reflective polarizer **116** (e.g., a linear, reflective polarizer). The lens arrangement **102** can also comprise an optical retarder **120**. The optical retarder **120** (e.g., a quarter-wave plate) is between the first mirror M-1 and the second mirror M-2. Light emitted from the display **104** is transmitted by the partial reflector **112**, the first mirror M-1 (e.g., half the light is transmitted through the partial reflector **112**); reflected off the second mirror M-2; passed back to toward the first mirror M-1; reflected by the first mirror M-1; and then transmitted through the reflective polarizer **116**, the second mirror M-2, to the eye box **108**. An eye of the user is positioned at the eye box **108**. By folding an optical bath onto itself, a form factor of the artificial-reality display **100** can be reduced (e.g., a distance between the display **104** and the eye box **108** can be reduced in the z dimension).

(19) The first mirror M-1 can be part of a first lens **124-1**. In some configurations, the lens arrangement **102** comprises a second lens **124-2**. The first lens **124-1** and/or the second lens **124-2** are configured to focus light to the eye box **108**. In some embodiments, the reflective polarizer **116** is part of the second lens **124-2**.

(20) FIG. 2 depicts a schematic of a display path **204** of an embodiment of an artificial-reality display with a lens arrangement **102**. The display path **204** is an optical path of light transmitted from the display **104** to the eye box **108**. FIG. 2 depicts polarization of light along the display path **204**. The optical path is along the z-axis. The x-axis is not shown because the schematic in FIG. 3 is for describing light polarization as light bounces in the lens arrangement **102** and is not meant to show a change along the x axis.

(21) The display **104** emits light. Light from the display **104** is transmitted to the partial reflector **112**. Light between the display **104** and the partial reflector **112** is left-handed, circularly-polarized (LHC) light. Light between the display **104** and the partial reflector **112** could be left-handed,

circularly-polarized light because light could be emitted from the display **104** as left-handed, circularly-polarized light, or light from the display **104** could be linearly polarized and changed to circularly-polarized light by a waveplate **208**. A waveplate, sometimes also referred to as a retarder, is an optical element that changes polarization of light traveling through the optical element. The waveplate **208** could be a quarter-wave plate with a fast axis oriented at 45 degrees with respect to the x-axis. Light linearly polarized along the x-axis, or horizontally polarized, is sometimes referred to as p-polarized light. Light linearly polarized along the y-axis, or vertically polarized, is sometimes referred to as s-polarized light.

(22) Some of the light from the display **104** is transmitted through the partial reflector **112**. In some embodiments, the partial reflector **112** has a transmission of 50% and reflectance of 50% (e.g., the partial reflector **112** has transmission of 50%+/-2, 5, 10, or 15%). Thus 50% of the light from the display **104** is reflected by the partial reflector **112**, and lost.

(23) Light transmitted by the partial reflector **112** is left-handed, circularly-polarized light and is transmitted to the optical retarder **120**. The optical retarder **120** has a fast axis oriented at 135 degrees with respect to the x-axis. Left-handed, circularly-polarized light passing through the optical retarder **120** in a positive z-direction will be changed to s-polarized light.

(24) The reflective polarizer **116** has a transmission axis aligned with the x-axis (which passes p-polarized light). Thus s-polarized light will be reflected from the reflective polarizer **116** and pass through the optical retarder **120** in a negative z-direction. S-polarized light passing through the optical retarder **120** in the negative z-direction will be changed to left-handed, circularly-polarized light. After passing through the optical retarder **120** a second time, and traveling in the negative z-direction, light will reflect from the partial reflector **112** (and 50% will be transmitted in the negative z-direction and be lost; thus a total of 75% of light from the display **104** is lost by using the partial reflector **112**).

(25) Light that is left-handed, circularly-polarized light traveling in the negative z-direction will be reflected from the partial reflector **112** as right-handed, circularly-polarized (RHC) light propagating in the positive z-direction. The right-handed, circularly-polarized light reflected from the partial reflector **112** passes through the optical retarder **120**, a third time (a second time in the positive z-direction), and is converted to p-polarized light. P-polarized light from the optical retarder **120** is passed by the reflective polarizer **116** because the reflective polarizer **116** passes p-polarized light. Passed light from the reflective polarizer **116** is transmitted to the eye box **108**.

(26) FIG. 3 is a picture of ghost images from a camera used for eye tracking with a lens arrangement **102** having a first mirror M-1 and a second mirror M-2. The picture is of black and white squares. The camera is configured to track a position of an eye of a user of the artificial-reality system. The first mirror M-1 and/or the second mirror M-2 provide reflections that produce ghost images for the camera. Light traveling along an optical path that produces a ghost image follows what is called a ghost path. Light following a ghost path produces an image that can interfere with an image of the eye, which interfere with tracking the eye.

(27) FIG. 4 illustrates a ray tracing diagram of an embodiment of an artificial-reality system having a camera **402** used for eye tracking with the lens arrangement **102**. Embodiments of ghost paths **404** and a display path **204** are shown. Also shown is an embodiment of a tracking path **408**. The tracking path **408** is an optical path of light used for imaging an eye **412** of a user. Light traveling along the tracking path **408** travels from the eye **412**, through the lens arrangement, and directed by a reflector **416** to the camera **402**. The camera **402** comprises an image sensor **424**. The image sensor **424** acquires images of the eye.

(28) For eye tracking in a system without the lens arrangement **102**, the reflector **416** can be a hot mirror. A hot mirror is a mirror that reflects light of longer wavelengths (e.g., infrared light) and passes light of shorter wavelengths (e.g., visible light). If a hot mirror is used as the reflector **416** with the lens arrangement **102**, light is able to bounce between the reflector **416** and the first mirror M-1 (e.g., a first ghost path **404-1**); and/or light is able to bounce between the first mirror M-1 and

the second mirror M-2 (a second ghost path **404-2**).

(29) One possible solution to reduce ghost images is to use have the first mirror M-1 be a cold mirror. FIG. 5 illustrates a ray tracing diagram of an embodiment of the first mirror M-1 having a dielectric coating. The dielectric coating is used to have the first mirror M-1 be a cold mirror. The cold mirror passes light of longer wavelengths (e.g., infrared) and reflects light of shorter wavelengths (e.g., visible light). Thus the first mirror M-1 could have a reflectance of 50%,  $\pm 5\%$  for light having a wavelength equal to or less than 750 nm (and/or equal to or greater than 450 nm); and the cold mirror could have a reflectance of less than 1%, 5% or 10% for light having a wavelength equal to or greater than 800 nm (and/or equal to or less than 2.5 microns). The image sensor **424** images light equal to or greater than 800 nm and/or equal to or less than 2.5 microns. Thus the first mirror M-1 in FIG. 5 will not cause ghost images by spurious reflections.

(30) The reflector **416** is a hot mirror **516**. The dielectric coating prevents light from transmitting along the first ghost path **404-1** to reach the camera **402** because light from the eye **412** of a user, reflecting off the hot mirror **516** would not be reflected from first mirror M-1 back toward the hot mirror **516**. The dielectric coating prevents the second ghost path **404-2** because infrared light reflected from the eye is not reflected by the first mirror M-1 back toward the second mirror M-2.

(31) Another possible solution to reduce ghost images is to use a reflective, circular polarizer. FIG. 6 illustrates a ray tracing diagram of an embodiment of an eye-tracking system with circular polarizer **616** used as reflector **416** to direct light to the camera **402** for eye tracking. The circular polarizer **616** is a reflective, circular polarizer. In some embodiments, the circular polarizer **616** is a narrow band polarizer (e.g., having a reflection bandwidth equal to or less than 20, 30, 40, or 70 nm, and/or equal to or greater than 5, 10, or 20 nm). The circular polarizer **616** can have a short-wavelength limit. A short-wavelength limit is the shortest wavelength that is reflected at half the maximum reflectance of the circular polarizer **616**. A high-wavelength limit is the longest wavelength that is reflected at half the maximum reflectance of the circular-polarizer **616**.

Reflection bandwidth of the circular polarizer **616** is measured between the short-wavelength limit and the high-wavelength limit. In some embodiments, the circular polarizer **616** is a high-pass reflector, meaning high frequencies (low wavelengths) are passed and low frequencies (high wavelengths) are reflected (e.g., a high-pass reflector can have a short-wavelength limit and no high-wavelength limit). In some embodiments, reflection of the circular polarizer **616** is matched with the display **104**. For example, the short-wavelength limit of the circular polarizer **616** is configured to be at a longer wavelength than a spectrum of light of the display **104** (e.g., where the spectrum of light of the display **104** is measured as full-width at half-maximum of light emitted by the display **104**). In some embodiments, the short-wavelength limit of the circular polarizer **616** is longer than the longest wavelength of light emitted by the display **104**. The short-wavelength limit of the circular polarizer **616** can be equal to or greater than 680, 700, 720, or 750 nm and/or equal to or less than 800, 1000, or 2000 nm. The circular polarizer **616** reflects one circular polarization (e.g., right-handed, circularly-polarized light) and passes an orthogonal circular polarization (e.g., left-handed, circularly-polarized light). Unlike reflection from a metallic mirror where circular polarization changes handedness, reflection from the circular polarizer **616** maintains handedness of the circular polarization upon reflection (e.g., the circular polarizer **616** is made using a cholesteric liquid crystal).

(32) Light from an eye **412** of the user traveling away from the eye **412** passes through the second mirror M-2 (e.g., a reflective, linear polarizer with a transmission axis aligned with the x-axis) and is linearly polarized (e.g., p-polarized). After passing through the second mirror M-2, light passes through the optical retarder **120**. The optical retarder **120** is a quarter-wave plate having a fast axis oriented at  $135^\circ$  from the x-axis. P-polarized light passing through the optical retarder **120** will be converted to right-handed, circularly-polarized light (RHC). The right-handed, circularly-polarized light will be reflected by the circular polarizer **616**.

(33) Light is not transmitted to the camera **402** on the first ghost path **404-1** because light reflecting

from the circular polarizer **616** back toward the first mirror M-1 (which is RHC) is reflected by the first mirror M-1 as left-handed, circularly-polarized light and propagates towards the circular polarizer **616**. The left-handed, circularly-polarized light then passes through the circular polarizer **616** and is not deflected by the circular polarizer **616** toward the camera **402**.

(34) Light is not transmitted to the camera **402** on the second ghost path **404-2** because light reflecting from the first mirror M-1 back toward the optical retarder **120** will be converted to left-handed, circularly-polarized light. Left-handed, circularly-polarized light traveling in the z-direction will be converted to p-polarized light; reflected by the second mirror M-2 to propagate in the negative z-direction; pass back through the optical retarder **120**; and be converted to left-handed, circularly-polarized light traveling toward first mirror M-1. Left-handed, circularly-polarized light traveling toward the first mirror M-1 will be partially transmitted and partially reflected. Partially reflected light will be transmitted back through the reflective polarizer **116** (M-2) as described in FIG. 2. Partially transmitted light will be transmitted as left-handed, circularly-polarized light, which will pass through the circular polarizer **616** and not be directed toward the camera **402**.

(35) Another possible solution to reduce ghost images is to use a waveplate. FIG. 7 illustrates a ray tracing diagram of an embodiment of a system **700** for eye tracking with folded optics. The system **700** comprises the lens arrangement **102** for focusing light from the display **104** to the eye **412** of a user (e.g., to the eye box **108**); the camera **402**; a reflector **416**, wherein the reflector **416** is configured to direct a portion of light to the camera **402** after the portion of light passes through the lens arrangement **102**; a first waveplate **704-1**, wherein the first waveplate **704-1** is between the lens arrangement **102** and the reflector **416**; a second waveplate **704-2**, wherein the second waveplate **704-2** is between the reflector **416** and the camera **402**; and a polarizer **708**, wherein the polarizer **708** is between the second waveplate **704-2** and the camera **402**. The lens arrangement **102** comprises the first mirror M-1 and the second mirror M-2.

(36) The reflector **416** can be a hot mirror (e.g., reflecting infrared light and passing visible light). By adding the first waveplate **704-1**, the second waveplate **704-2**, and the polarizer **708** to the configuration shown in FIG. 4, ghost paths **404** can be reduced and/or eliminated.

(37) In some embodiments, the first waveplate **704-1** is a first quarter-wave plate, the second waveplate **704-2** is a second quarter-wave plate, and/or the polarizer **708** is a linear polarizer (e.g., an absorptive polarizer). The first quarter-wave plate is clocked by 90 degrees from the second quarter-wave plate. The first quarter-wave plate is clocked by 90 degrees from the optical retarder **120** (e.g., the first waveplate **704-1** has a fast axis oriented by about 45 degrees with respect to the x-axis). In some embodiments, the first waveplate **704-1** is the waveplate **208** in FIG. 2. The polarizer **708** is an absorptive, linear polarizer with a transmission axis aligned horizontally, to pass p-polarized light.

(38) Light from the display **104** travels along the display path **204** to the eye box **108**. Light reflections off the eye **412** travel along the tracking path **408** to the camera **402** and are imaged by the image sensor **424**. Light reflected from the eye **412** traveling along a ghost path **404** is attenuated by the polarizer **708**.

(39) FIG. 8 depicts a schematic of optical paths of an eye-tracking system **700**. In FIG. 8, polarization states of light along the tracking path **408**, the first ghost path **404-1**, and the second ghost path **404-2** are shown. The y-axis (vertical axis) is into and out of the page.

(40) For the tracking path **408**, light is transmitted through the second mirror M-2 and p-polarized light because the second mirror M-2 is a linear, reflective polarizer **116** with a horizontal transmission axis. P-polarized light traveling from the second mirror M-2 and through the optical retarder **120** will be changed to right-handed, circularly-polarized (RHC) light. After passing through the optical retarder **120**, light will pass through the first mirror M-1, without changing polarization. After passing through the first mirror M-1, light passes through the first waveplate **704-1** and is changed back to p-polarized light. P-polarized light is then reflected as p-polarized



light by the reflector **416**, passes back through the first waveplate **704-1**, and is converted back to right-handed circularly-polarized light. After passing through the first waveplate **704-1** a second time, light passes through the second waveplate **704-2** and is converted to p-polarized light. The p-polarized light, after passing through the second waveplate **704-2**, passes through the polarizer **708** and to the camera **402**, because p-polarized light is aligned with the transmission axis of the polarizer **708**.

(41) For the first ghost path **404-1**, light is transmitted through the second mirror M-2 and p-polarized because the second mirror M-2 is a linear, reflective polarizer **116** with a horizontal transmission axis. P-polarized light traveling from the second mirror M-2 and through the optical retarder **120** will be changed to right-handed, circularly-polarized light. After passing through the optical retarder **120**, light will pass through the first mirror M-1, without changing polarization. After passing through the first mirror M-1, light passes through the first waveplate **704-1** and is changed back to p-polarized light. P-polarized light is then reflected as p-polarized light by the reflector **416**, passed back through the first waveplate **704-1** and converted back into right-handed circularly-polarized light. After passing through the first waveplate **704-1** a second time, light is transmitted back toward the first mirror M-1 and reflects off the first mirror M-1 as left-handed, circularly-polarized (LHC) light. Left-handed, circularly-polarized light traveling from the first mirror M-1 toward the first waveplate **704-1** will be changed to s-polarized light by passing through the first waveplate **704-1**. S-polarized light is then reflected as s-polarized light by the reflector **416**, passed back through the first waveplate **704-1**, and converted to left-handed circularly-polarized light. Left-handed, circularly-polarized light passing through the second waveplate **704-2** toward the camera **402** will be changed to s-polarized light and blocked by the polarizer **708** because the polarizer **708** transmits p-polarized light and blocks s-polarized light.

(42) For the second ghost path **404-2**, light is transmitted through the second mirror M-2 and p-polarized because the second mirror M-2 is a linear, reflective polarizer **116** with a horizontal transmission axis. P-polarized light traveling from the second mirror M-2 and through the optical retarder **120** will be changed to right-handed, circularly-polarized light. After passing through the optical retarder **120**, light is reflected as left-handed circularly-polarized light by the first mirror M-1, back toward the second mirror M-2. Left-handed, circularly polarized light traveling in the z-direction changes into s-polarized light after passing through the optical retarder **120**, is reflected by the second mirror M-2 as s-polarized light, and converted back into left-handed, circularly-polarized light traveling in a negative z-direction. Left-handed, circularly-polarized light passes through the first mirror M-1 as left-handed, circularly-polarized light. Left-handed, circularly-polarized traveling from the first mirror toward the first waveplate **704-1** will be changed to s-polarized light by passing through the first waveplate **704-1**. S-polarized light is then reflected as s-polarized light by the reflector **416**, passed back through the first waveplate **704-1**, and converted to left-handed circularly-polarized light. Left-handed, circularly-polarized light passing through the second waveplate **704-2** toward the camera **402** will be changed to s-polarized light and blocked by the polarizer **708** because the polarizer **708** transmits p-polarized light and blocks s-polarized light. Thus the first waveplate **704-1**, the second waveplate **704-2**, and the polarizer **708** block ghost paths **404** while allowing light from the tracking path **408** to enter the camera **402**.

(43) FIG. **9** is a picture of the *Narcissus* effect. The *Narcissus* effect is the camera “seeing” itself based on reflections from the camera. To reduce the *Narcissus* effect, a third waveplate is placed between the polarizer **708** and the camera **402**.

(44) FIG. **10** depicts a schematic of an optical path of an eye-tracking system for reducing the *Narcissus* effect. To reduce the *Narcissus* effect, a third waveplate **704-3** is placed between the polarizer **708** and the camera **402**. Referring back to FIG. **8**, p-polarized light travels from the second waveplate **704-2** toward the polarizer **708** and is passed by the polarizer **708**. FIG. **10** depicts the polarizer **708** and the camera **402** from FIG. **8**. FIG. **10** further depicts the third waveplate **704-3** between the polarizer **708** and the camera **402**. The third waveplate **704-3** has a

fast axis oriented  $\pm 45$  degrees with respect to the transmission axis of the polarizer **708**. Taking the example of the fast axis being clocked similarly as the first waveplate **704-1**, p-polarized light is changed into right-handed, circularly-polarized light by passing through the third waveplate **704-3**. A first part of the right-handed, circularly-polarized light will pass through camera optics and be detected by the image sensor **424**. A second part of the right-handed, circularly-polarized light transmitted from the third waveplate **704-3** will be reflected by camera optics (e.g., a surface of a lens of the camera **402**) back toward the third waveplate **704-3** as left-handed, circularly-polarized light. Reflected left-handed, circularly-polarized light from the camera optics changes to s-polarized light by passing through the third waveplate **704-3** and blocked by the polarizer **708**. Thus reflections from camera optics are attenuated by inserting the third waveplate **704-3** between the polarizer **708** and the camera **402**.

(45) FIG. **11** illustrates an embodiment of a flowchart of a process **1100** for using an eye-tracking system with the lens arrangement **102**. Process **1100** begins in step **1104** with directing a portion of light to the camera **402**, using a reflector **416**, after the portion of light passes through the lens arrangement **102** and the first waveplate **704-1**, wherein the lens arrangement **102** comprises the first mirror M-1 and the second mirror M-2. The portion of light can be a reflection from an eye of a user. The portion of light is passed through the first waveplate, step **1108**, after directing the portion of light with the reflector **416**. In step **1112**, the portion of light is passed through the second waveplate **704-2**, wherein the second waveplate **704-2** is between the reflector **416** and the camera **402**. After passing through the second waveplate **704-2**, the portion of light is passed through the polarizer **708**, step **1116**. After passing through the polarizer **708**, the portion of light is detected by the camera **402** (e.g. using the image sensor **424**), step **1120**. The camera **402** is configured to capture images of the eye of the user; and a controller is configured to track a position of the eye of the user base on images of the eye captured by the camera. In some configurations, the process **100** further comprises passing the portion of light through the third waveplate **704-3**, after passing the portion of light through the second waveplate **704-2** and before detecting the portion of light with the camera **402**.

(46) Embodiments of the invention may include or be implemented in conjunction with an artificial reality system. Artificial reality is a form of reality that has been adjusted in some manner before presentation to a user, which may include, e.g., a virtual reality (VR), an augmented reality (AR), a mixed reality (MR), a hybrid reality, or some combination and/or derivatives thereof. Artificial reality content may include completely generated content or generated content combined with captured (e.g., real-world) content. The artificial reality content may include video, audio, haptic feedback, or some combination thereof, and any of which may be presented in a single channel or in multiple channels (such as stereo video that produces a three-dimensional effect to the viewer). Additionally, in some embodiments, artificial reality may also be associated with applications, products, accessories, services, or some combination thereof, that are used to, e.g., create content in an artificial reality and/or are otherwise used in (e.g., perform activities in) an artificial reality. The artificial reality system that provides the artificial reality content may be implemented on various platforms, including a head-mounted display (HMD) connected to a host computer system, a standalone HMD, a mobile device or computing system, or any other hardware platform capable of providing artificial reality content to one or more viewers.

(47) The foregoing description of the embodiments of the disclosure has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.

(48) Some portions of this description describe the embodiments of the disclosure in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while

described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, and/or hardware.

(49) Steps, operations, or processes described may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In some embodiments, a software module is implemented with a computer program product comprising a computer-readable medium containing computer program code, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described.

(50) Embodiments of the disclosure may also relate to an apparatus for performing the operations described. The apparatus may be specially constructed for the required purposes, and/or it may comprise a general-purpose computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory, tangible computer readable storage medium, or any type of media suitable for storing electronic instructions, which may be coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

(51) Embodiments of the disclosure may also relate to a product that is produced by a computing process described herein. Such a product may comprise information resulting from a computing process, where the information is stored on a non-transitory, tangible computer readable storage medium and may include any embodiment of a computer program product or other data combination described herein.

(52) The language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the disclosure be limited not by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments is intended to be illustrative, but not limiting, of the scope of the disclosure, which is set forth in the following claims.

## Claims

1. A system comprising: a lens arrangement for projecting light from a display to an eye of a user, the lens arrangement comprising: a first mirror; and a second mirror; a camera; and a reflector configured to direct a portion of light to the camera after the portion of light passes through the lens arrangement, wherein the reflector is a reflective, circular polarizer that passes circularly-polarized light of one handedness and reflects circularly-polarized light of an orthogonal handedness.
2. The system of claim 1, wherein: the reflector has a short-wavelength limit that is greater than 700 nm; and the short-wavelength limit is a shortest wavelength that is reflected at half a maximum reflectance of the reflector.
3. The system of claim 1, wherein the portion of light is reflected by the eye of the user.
4. The system of claim 3, wherein the camera is configured to track a position of the eye based on receiving the portion of light.
5. The system of claim 1, wherein the reflector is a narrow band polarizer characterized by a reflection bandwidth equal to or less than 70 nm.
6. The system of claim 1, wherein the reflector is a high-pass reflector configured to reflect light with wavelengths greater than a threshold value.
7. The system of claim 1, wherein the reflector is configured to reflect the circularly-polarized light of the orthogonal handedness and maintain the orthogonal handedness in the reflected circularly-polarized light.

8. The system of claim 1, wherein the reflector includes a cholesteric liquid crystal.
  9. The system of claim 1, wherein the second mirror includes a reflective, linear polarizer.
  10. The system of claim 9, wherein the lens arrangement includes an optical retarder between the first mirror and the second mirror.
  11. The system of claim 10, wherein the optical retarder includes a quarter-wave plate.
  12. The system of claim 1, wherein the first mirror includes a partial reflector.
  13. The system of claim 1, wherein the first mirror is on a curved surface of a lens.
  14. The system of claim 1, wherein the reflector is between the display and the lens arrangement.
  15. The system of claim 1, wherein: the lens arrangement includes two or more lenses; and at least one of the first mirror or the second mirror is formed on a surface of a lens of the two or more lenses.
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